

Design a Web Crawler - Step 4: Wrap Up

we covered:

1. Characteristics of a Good Web Crawler

- **Scalability** → Handle billions of pages efficiently
- **Politeness** → Avoid overwhelming websites
- **Extensibility** → Easily support new content types
- **Robustness** → Handle failures, bad data, and edge cases

2. High-Level Design

- Proposed system architecture
- Discussed all major components like:
 - URL Frontier
 - HTML Downloader
 - Content Parser
 - Storage systems
 - Filtering mechanisms

Key Insight

Building a **large-scale web crawler is complex** because:

- The web is **massive (billions of pages)**
- Contains:
 - Duplicate content
 - Malicious traps
 - Dynamic content

Additional Important Talking Points

1. Server-Side Rendering

Problem

- Many websites use:
 - JavaScript
 - AJAX

Links are generated **dynamically**

Issue

- Simple HTML parsing **cannot extract dynamic links**

Solution

- Perform **server-side rendering (dynamic rendering)** before parsing

2. Filter Out Unwanted Pages

Problem

- Limited:
 - Storage
 - Crawl capacity

Solution

- Use **anti-spam filtering systems**

Goal

- Remove:
 - Low-quality pages
 - Spam content

3. Database Replication and Sharding

Why Needed?

- Massive data storage requirements

Techniques

- **Replication**
 - Improves availability and reliability
- **Sharding**
 - Distributes data across multiple databases

4. Horizontal Scaling

Concept

- Add more servers to scale system

Key Requirement

- Keep servers **stateless**

Benefit

- Easy to:
 - Scale up/down
 - Handle large workloads

5. Availability, Consistency, Reliability

Core System Design Concepts

- **Availability** → System always accessible
- **Consistency** → Data remains correct
- **Reliability** → System works without failure

6. Analytics

Importance

- Data helps improve system performance

Use Cases

- Crawl efficiency analysis
- Content quality evaluation
- System optimization

Final Takeaway

A production-grade web crawler must:

- Scale to **billions of pages**
- Handle **dynamic + noisy web content**

- Maintain **high performance and reliability**
- Continuously improve using **analytics**

Conclusion:

Designing a web crawler is not just about crawling pages, but about building a **robust, scalable, and intelligent system** capable of handling real-world web complexity.